# Option Pricing: Motivation and First Mathematical Tools

# How the course materials fit together

**Slides**

The in-class : core ideas, examples, and formulas.

**Lecture notes**

The full reference: definitions, derivations, proofs, and worked examples.

**Exercises and HW**

The practice layer: calculations, code...

# Homework and final grade

## Homework rhythm

- Homework is assigned **weekly**; you have **two weeks** to complete each assignment. Brightspace gives the official deadline and submission instructions.
- **Practice assignments:** mathematical reasoning and calculations.
- **Numerical assignments:** simulation or approximation, with code and interpretation.
- The **lowest homework score is dropped**.

## Two grading routes

| | **Grade 1** | **Grade 2** |
|---|---|---|
| Homework average | 50% | 60% |
| Midterm exam | 20% | 0% |
| Final exam | 30% | 40% |

$$\text{Final Grade} = \max\big(\text{Grade 1}, \text{Grade 2}\big)$$

The more favorable calculation is used automatically.

## Letter-grade scale

| **F** | **D** | **D+** | **C-** | **C** | **C+** | **B-** | **B** | **B+** | **A-** | **A** |
|---|---|---|---|---|---|---|---|---|---|---|
| $< 62.45$ | 62.45 | 66.45 | 69.45 | 72.45 | 76.45 | 79.45 | 82.45 | 86.45 | 89.45 | 94.45 |

*Each value indicates the minimum percentage required for that letter grade.*

# References (see Brightspaces)

**Main references**

- S. Shreve, *Stochastic Calculus for Finance II*, Chapters 1–2.
- I. Karatzas and S. Shreve, *Brownian Motion and Stochastic Calculus*, Chapter 1.
- S. Shreve, *Stochastic Calculus for Finance I*, Chapters 1–2 for discrete-time pricing intuition.

# Roadmap

Motivation, replication, and pricing

Probability toolbox

Information and conditional expectation

Processes and martingales

Short differential-calculus review

Motivation, replication, and pricing

Contracts and time value

The casino reveals the role of replication

A one-period financial market

# Derivatives reshape future financial risk

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**The pricing question**

What amount paid **today** is consistent with a possibly random **future payoff**?

# One option introduces most of the vocabulary

Suppose a one-year call option on a stock has strike $K = 100$:

$$X = (S_T - K)^+ = (S_T - 100)^+.$$

- $S$ is the **underlying**; $T$ is the **maturity**.

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- The **long** pays the option price today and receives $X$ at $T$.

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- $X$ is the **payoff**: its rule is known today and its value is realized at $T$.
- The **long** pays the option price today and receives $X$ at $T$.
- The **short** receives the price today and owes $X$ at $T$.

If $S_T = 125$, then $X = 25$; if $S_T = 90$, then $X = 0$.

# A dollar today and a dollar tomorrow are different

At a constant continuously compounded rate $r$, the bank account is $B_t = e^{rt}$.

## Discounting

A certain payment of $C$ at time $T$ has value

$$Ce^{-rT}$$

today, because this amount invested in the bank account produces exactly $C$ at $T$.

## Example

If $C = 100$, $r = 5\%$, and $T = 1$, the present value is

$$100e^{-0.05} \approx 95.12.$$

# Historical expectation is a natural first guess

I sell a contract with payoff

$$X = \begin{cases} \$100, & \text{if red occurs,} \\ \$0, & \text{if black occurs.} \end{cases}$$

Under the historical probability measure $\mathbb{P}$,

$$\mathbb{P}(\text{red}) = 0.99, \qquad \mathbb{P}(\text{black}) = 0.01,$$

so

$$\mathbb{E}^{\mathbb{P}}[X] = 0.99 \times \$100 + 0.01 \times \$0 = \$99.$$

**Pricing question**

Is \$99 the arbitrage-free price? **No.** Historical probabilities describe likelihoods; they do not by themselves price a replicable claim.

# A $50 bet reproduces the contract state by state

What payoff does $50 placed on red produce?

| | Red | Black |
|---|---|---|
| Contract | $100 | $0 |
| $50 bet on red | $2 \times \$50 = \$100$ | $0 |

same terminal payoff in every state $\Longrightarrow$ replication cost $= \$50$.

**Takeaway**

**PRICE = COST OF REPLICATION**

# No arbitrage forces the price to be $50

$$C_0 = \$50, \qquad \text{not} \qquad \mathbb{E}^{\mathbb{P}}[X] = \$99.$$

**If the contract costs more than** $50

- sell the contract;
- buy the $50 hedge;
- keep the difference with no terminal risk.

**If it costs less than** $50

- buy the contract;
- sell the replicating bet;
- keep the reverse difference.

**Transition**

Historical probabilities describe likelihoods. Which probabilities represent prices consistently with traded assets?

# A one-period stock model makes replication visible

Assume a zero interest rate. A stock is worth $S_0 = 50$ today and

$$S_1 = \begin{cases} 60 & \text{up}, \\ 40 & \text{down}. \end{cases}$$

A call with strike $K = 50$ pays

$$X = (S_1 - 50)^+ = \begin{cases} 10 & \text{up}, \\ 0 & \text{down}. \end{cases}$$

Can stock and cash reproduce both payoffs?

# Worked example: replication determines the call price

Hold $\Delta$ shares and $b$ dollars in cash. Matching both terminal payoffs gives

$$60\Delta + b = 10, \qquad 40\Delta + b = 0.$$

Therefore

$$\Delta = \tfrac{1}{2}, \qquad b = -20, \qquad V_0 = \Delta S_0 + b = 5.$$

**Law of one price**

The call and its replicating portfolio have the same payoff in every state, so both must cost 5 today.

# Risk-neutral probabilities encode the same restriction

Choose $q = \mathbb{Q}(\text{up})$ so that the stock has no drift at zero interest:

$$50 = q \times 60 + (1 - q) \times 40 \quad \Longrightarrow \quad q = \tfrac{1}{2}.$$

Then

$$V_0 = \mathbb{E}^{\mathbb{Q}}[X] = \tfrac{1}{2} \times 10 + \tfrac{1}{2} \times 0 = 5.$$

**Interpretation**

The physical probability may differ from $q$. Replication determines the price first; the risk-neutral expectation is its mathematical representation.

# Student exercise: replicate a similar call

Assume a zero interest rate. A stock is worth $S_0 = 100$ and

$$S_1 = 125 \text{ or } 75.$$

A call with strike $K = 100$ pays 25 or 0.

1. Write the two replication equations for $(\Delta, b)$.
2. Find the no-arbitrage call price.
3. Find the risk-neutral up probability.
4. Explain why the physical up probability is unnecessary.

**Hint**

Subtract the two payoff equations before solving for cash.

# Solution: replication and risk-neutral probability

Matching the two terminal payoffs gives

$$125\Delta + b = 25, \qquad 75\Delta + b = 0.$$

Subtracting the equations,

$$\Delta = \frac{25}{125 - 75} = \frac{1}{2}, \qquad b = -37.5.$$

Hence the replication cost is

$$V_0 = \Delta S_0 + b = \frac{1}{2} \times 100 - 37.5 = 12.5.$$

The risk-neutral probability solves

$$100 = 125q + 75(1 - q), \qquad q = \frac{1}{2},$$

and therefore $\mathbb{E}^{\mathbb{Q}}[X] = 12.5$. The physical up probability is not needed.

# Replication motivates the tools used throughout the course

For a replicable payoff $X$, an arbitrage-free model represents its value as

$$V_t = B_t \, \mathbb{E}^{\mathbb{Q}} \left[ \frac{X}{B_T} \,\middle|\, \mathcal{F}_t \right].$$

To understand this expression, we now build a probability toolbox in layers:

distribution $\longrightarrow$ expectation $\longrightarrow$ information $\longrightarrow$ conditional prediction $\longrightarrow$ martingale.

# Probability toolbox

## Probability toolbox

- Random variables, distributions, CDFs, and densities
- Expectation, variance, and dependence
- Sample averages, LLN, and CLT
- Characteristic functions and joint distributions

# Introductory example: a payoff is a random variable

Suppose the terminal stock price has the following physical distribution:

| $S_T$ | 80 | 100 | 130 |
|---|---|---|---|
| $\mathbb{P}$ | 0.20 | 0.50 | 0.30 |

For a call with strike $K = 100$, $X = (S_T - 100)^+$:

| $S_T$ | 80 | 100 | 130 |
|---|---|---|---|
| $X$ | 0 | 0 | 30 |

The payoff rule maps each state to a number; the probabilities of those numbers form the **distribution of** $X$.

# Definitions: random variable and distribution

## Random variable and its law

A random variable is a measurable map

$$X : (\Omega, \mathcal{F}) \longrightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R})).$$

It assigns a numerical outcome to every state. Its distribution, or law, is

$$\mathbb{P}_X(A) = \mathbb{P}(X \in A), \qquad A \in \mathcal{B}(\mathbb{R}).$$

The random variable is the state-to-number rule; its distribution records how likely its possible values are.

# The CDF accumulates probability from left to right

## Cumulative distribution function

$$F_X(x) = \mathbb{P}(X \le x).$$

Every CDF is nondecreasing, right-continuous, and satisfies

$$\lim_{x \to -\infty} F_X(x) = 0, \qquad \lim_{x \to \infty} F_X(x) = 1.$$

Probabilities of intervals follow from differences:

$$\mathbb{P}(a < X \le b) = F_X(b) - F_X(a).$$

**Do not confuse the objects**

The **distribution** is a probability measure; the **CDF** is its cumulative function; a **density** is a derivative when one exists.

# Worked example: the exponential CDF comes from its density

Let $X \sim \mathrm{Exp}(\lambda)$ with

$$f_X(x) = \lambda e^{-\lambda x} \mathbf{1}_{\{x \geq 0\}}.$$

Then, for $x \geq 0$,

$$F_X(x) = \int_{-\infty}^{x} f_X(u)\, du = \int_0^x \lambda e^{-\lambda u}\, du = 1 - e^{-\lambda x}.$$

Thus

$$\mathbb{P}(a < X \leq b) = e^{-\lambda a} - e^{-\lambda b}, \qquad 0 \leq a < b.$$

When differentiability holds,

$$f_X(x) = F_X'(x).$$

# Definition: expectation is a probability-weighted average

**Expectation**

For an integrable random variable $X$,

$$\mathbb{E}[X] = \sum_x x\,\mathbb{P}(X = x) \quad \text{or} \quad \mathbb{E}[X] = \int_{\mathbb{R}} x f_X(x)\,dx.$$

Integrable means $\mathbb{E}[|X|] < \infty$, written $X \in L^1$.

Expectation uses a distribution to summarize the long-run or model-based average. It has the same financial unit as $X$.

# Functions of a random variable can be averaged directly

## Averaging a transformation

For measurable $g$, whenever the expectation exists,

$$\mathbb{E}[g(X)] = \sum_x g(x)\mathbb{P}(X = x) \quad \text{or} \quad \mathbb{E}[g(X)] = \int_{\mathbb{R}} g(x) f_X(x)\, dx.$$

## Indicator payoff

For $X = \mathbf{1}_A$,

$$\mathbb{E}[X] = \mathbb{P}(A).$$

This is why a digital payoff is naturally connected to an event probability.

# Proposition: expectation is linear and respects order

For $X, Y \in L^1$ and $a, b \in \mathbb{R}$,

**Fundamental properties**

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y],$$
$$X \geq 0 \Longrightarrow \mathbb{E}[X] \geq 0,$$
$$X \leq Y \Longrightarrow \mathbb{E}[X] \leq \mathbb{E}[Y],$$
$$|\mathbb{E}[X]| \leq \mathbb{E}[|X|].$$

Linearity does not require independence. Independence matters when variances and products are calculated.

# Variance measures dispersion around the mean

## Mean, second moment, and variance

$$\mu = \mathbb{E}[X], \qquad m_2 = \mathbb{E}[X^2], \qquad \mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2].$$

The standard deviation is $\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}$.

## Useful identity

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2.$$

Mean describes location; variance describes uncertainty around that location.

# Covariance links dependence to portfolio variance

## Covariance

$$\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])].$$

## Variance of a linear combination

$$\mathrm{Var}(aX + bY) = a^2 \,\mathrm{Var}(X) + b^2 \,\mathrm{Var}(Y) + 2ab \,\mathrm{Cov}(X, Y).$$

If $X$ and $Y$ are independent, then $\mathrm{Cov}(X, Y) = 0$.

This identity explains why dependence matters when risks are combined.

# Reference: classical means and variances to know

| Law of $X$ | Parameters | $\mathbb{E}[X]$ | $\mathrm{Var}(X)$ |
|---|---|---|---|
| Bernoulli | $p$ | $p$ | $p(1-p)$ |
| Binomial | $n, p$ | $np$ | $np(1-p)$ |
| Poisson | $\lambda$ | $\lambda$ | $\lambda$ |
| Uniform | $[a, b]$ | $(a+b)/2$ | $(b-a)^2/12$ |
| Exponential | rate $\lambda$ | $1/\lambda$ | $1/\lambda^2$ |
| Gaussian | $\mu, \sigma^2$ | $\mu$ | $\sigma^2$ |

The exercise sheet contains the detailed derivations; class time focuses on choosing the right representation.

# Example A: derive discrete moments together

Let $B_1, \dots, B_n$ be independent Bernoulli($p$) variables and set

$$N_n = \sum_{k=1}^{n} B_k.$$

1. Use $B_k^2 = B_k$ to find $\mathbb{E}[B_k]$ and $\mathrm{Var}(B_k)$.
2. Use linearity to find $\mathbb{E}[N_n]$.
3. Where is independence used to find $\mathrm{Var}(N_n)$?

**Progressive hint**

Write the variance of the sum using covariance terms before setting them to zero.

# Solution: moments of a Bernoulli sum

For $B_k \sim \text{Bernoulli}(p)$,

$$B_k^2 = B_k, \qquad \mathbb{E}[B_k] = p, \qquad \text{Var}(B_k) = p - p^2 = p(1-p).$$

Linearity gives

$$\mathbb{E}[N_n] = \sum_{k=1}^{n} \mathbb{E}[B_k] = np.$$

Independence makes all cross-covariances vanish, so

$$\text{Var}(N_n) = \sum_{k=1}^{n} \text{Var}(B_k) = np(1-p).$$

**Where independence enters**

It is not needed for the mean; it is used when simplifying the variance of the sum.

# Exercise B: derive continuous moments independently

Choose one variable and justify every step.

1. If $U \sim \text{Unif}[a, b]$, compute $\mathbb{E}[U]$, $\mathbb{E}[U^2]$, and $\text{Var}(U)$.
2. If $T \sim \text{Exp}(\lambda)$, compute the same three quantities using integration by parts.
3. For $Z \sim \mathcal{N}(0, 1)$, use symmetry to identify which odd moments vanish.

**Extension**

For $X \sim \text{Poisson}(\lambda)$, start from $\mathbb{E}[X(X-1)]$ rather than expanding $X^2$ directly.

# Solution: classical moment calculations

| Law | $\mathbb{E}[X]$ | $\mathbb{E}[X^2]$ | $\mathrm{Var}(X)$ |
|---|---|---|---|
| $U \sim \mathrm{Unif}[a,b]$ | $\frac{a+b}{2}$ | $\frac{a^2+ab+b^2}{3}$ | $\frac{(b-a)^2}{12}$ |
| $T \sim \mathrm{Exp}(\lambda)$ | $\frac{1}{\lambda}$ | $\frac{2}{\lambda^2}$ | $\frac{1}{\lambda^2}$ |

For $Z \sim \mathcal{N}(0,1)$, symmetry gives

$$\mathbb{E}[Z^{2m+1}] = 0 \qquad (m \geq 0).$$

For $X \sim \mathrm{Poisson}(\lambda)$,

$$\mathbb{E}[X(X-1)] = \lambda^2 \quad \Longrightarrow \quad \mathbb{E}[X^2] = \lambda^2 + \lambda, \quad \mathrm{Var}(X) = \lambda.$$

# A sample summarizes repeated observations

Let $X_1, \dots, X_n$ be observations of the same quantity. Define

**Empirical mean and empirical variance**

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad v_n = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X}_n)^2.$$

$\overline{X}_n$ estimates the population mean and $v_n$ its variance. The unbiased sample variance is

$$s_n^2 = \frac{n}{n-1} v_n.$$

These statistics will later summarize samples of returns, losses, and simulated payoffs.

# The LLN finds the target; the CLT describes the error

Let $X_1, X_2, \ldots$ be i.i.d. with mean $\mu$.

## Law of Large Numbers

If $\mathbb{E}[|X_1|] < \infty$, then

$$\overline{X}_n \longrightarrow \mu = \mathbb{E}[X_1] \quad \text{in probability.}$$

If also $0 < \sigma^2 = \mathrm{Var}(X_1) < \infty$,

## Central Limit Theorem

$$\frac{\sqrt{n}\,(\overline{X}_n - \mu)}{\sigma} \xRightarrow[n\to\infty]{} \mathcal{N}(0,1) \quad \text{in law.}$$

The LLN says the error vanishes; the CLT says its typical size is $\sigma/\sqrt{n}$ and gives its approximate shape.

# Worked example: estimate a success probability

Let $X_i \sim \text{Bernoulli}(0.60)$ independently and let $\overline{X}_{100}$ be the success frequency in 100 trials.

$$\mathbb{E}[\overline{X}_{100}] = 0.60, \qquad \text{SD}(\overline{X}_{100}) = \sqrt{\frac{0.60(0.40)}{100}} \approx 0.049.$$

The LLN explains why the frequency should be near 0.60. The CLT gives

$$\mathbb{P}(0.50 \le \overline{X}_{100} \le 0.70) \approx \mathbb{P}(-2.04 \le Z \le 2.04) \approx 0.96.$$

**Why it matters**

Averages stabilize, but their residual sampling uncertainty is of order $1/\sqrt{n}$.

# Characteristic functions encode an entire distribution

## Characteristic function

$$\varphi_X(u) = \mathbb{E}[e^{iuX}], \qquad u \in \mathbb{R}.$$

- $|e^{iuX}| = 1$, so $\varphi_X$ always exists.
- The characteristic function uniquely determines the distribution.
- It turns probability calculations into Fourier calculations.
- Unlike a moment generating function, it does not require exponential moments.

This is a recognition tool for later work, not a full Fourier theory.

# Independent sums become products in Fourier space

If $X$ and $Y$ are independent,

**Product rule**

$$\varphi_{X+Y}(u) = \varphi_X(u)\varphi_Y(u).$$

For $Z \sim \mathcal{N}(0,1)$,

$$\varphi_Z(u) = e^{-u^2/2}.$$

Thus, for independent standard normals $Z_1, Z_2$,

$$\varphi_{Z_1+Z_2}(u) = e^{-u^2},$$

which identifies $Z_1 + Z_2 \sim \mathcal{N}(0,2)$.

# Joint distributions describe dependence between variables

Consider

| | $Y = 0$ | $Y = 1$ | $\mathbb{P}_X$ |
|---|---|---|---|
| $X = 0$ | 0.30 | 0.10 | 0.40 |
| $X = 1$ | 0.20 | 0.40 | 0.60 |
| $\mathbb{P}_Y$ | 0.50 | 0.50 | 1 |

- The four interior values form the **joint distribution**.
- Row and column sums are the **marginal distributions**.
- The joint law tells us how observing one variable changes what we know about the other.

This observation creates the need for information and conditional distributions.

# Information and conditional expectation

Information and conditional expectation

- Information first, formalism second
- Conditional distributions and conditional expectation
- Gaussian conditioning
- Properties used in dynamic pricing

# Introductory example: observing a signal reveals a partition

A fair die produces the full outcome $X(\omega) = \omega$. Instead of observing $X$, we observe only

$$Y = \mathbf{1}_{\{X \text{ is even}\}}.$$

Knowing $Y$ tells us which information set contains the outcome:

$$O = \{1, 3, 5\}, \qquad E = \{2, 4, 6\}.$$

- $Y$ distinguishes odd from even outcomes.
- $Y$ cannot distinguish 2 from 4 or 6.
- Observing $Y$ therefore reveals the partition $\{O, E\}$.

# The notation $\sigma(Y)$ records exactly what $Y$ reveals

After observing the parity signal $Y$, we can decide whether

$$\omega \in O, \qquad \omega \in E, \qquad \omega \in \Omega, \qquad \omega \in \emptyset.$$

We cannot decide whether $\omega = 2$, because $Y$ gives the same signal for 2, 4, and 6.

**Information generated by a random variable**

$$\sigma(Y) = \{\text{events whose occurrence can be determined from } Y\}.$$

For this example,

$$\sigma(Y) = \{\emptyset, O, E, \Omega\}.$$

Thus $\sigma(Y)$ is the information revealed by observing $Y$, written as a collection of observable events.

# A sigma-algebra formalizes observable yes-or-no questions

**$\sigma$-algebra**

A collection $\mathcal{G}$ of events is a $\sigma$-algebra if

- $\Omega \in \mathcal{G}$;
- $A \in \mathcal{G}$ implies $A^c \in \mathcal{G}$;
- $A_1, A_2, \ldots \in \mathcal{G}$ implies $\bigcup_n A_n \in \mathcal{G}$.

The closure rules ensure that observable questions can be combined consistently.

# Measurable means determined by the available information

## $\mathcal{G}$-measurable random variable

A random variable $Z$ is $\mathcal{G}$-measurable if

$$\{Z \in B\} \in \mathcal{G} \qquad \text{for every Borel set } B.$$

**Information interpretation**

"$Z$ is $\mathcal{G}$-measurable" means that the information in $\mathcal{G}$ is sufficient to determine $Z$.

In a finite model,

$$Z \text{ is } \sigma(Y)\text{-measurable} \quad \Longleftrightarrow \quad Z = g(Y) \text{ for some function } g.$$

Thus $Z$ must take the same value whenever $Y$ takes the same value.

# Worked example: measurable quantities are constant on signal sets

For the parity signal, a $\sigma(Y)$-measurable variable has the form

$$Z = a\mathbf{1}_O + b\mathbf{1}_E.$$

- $Z_1 = 10 \cdot \mathbf{1}_E$ is measurable: parity determines it.
- $Z_2 = X$ is not measurable: parity does not determine the die value.
- $Z_3 = (-1)^X$ is measurable: it is a function of parity.

**Finite-state test**

A variable is measurable with respect to the observed partition if and only if it is constant on every information set.

# Student exercise: what does a coarse price signal reveal?

A terminal stock price takes values 80, 100, or 130. The signal is

$$Y = \mathbf{1}_{\{S_T \geq 100\}}.$$

1. Write the partition generated by observing $Y$.
2. List the events in $\sigma(Y)$.
3. Is the call payoff $(S_T - 100)^+$ measurable with respect to $\sigma(Y)$?
4. Is the digital payoff $\mathbf{1}_{\{S_T \geq 100\}}$ measurable?

**Test**

Can two states with the same signal produce different values of the quantity?

## Solution: the signal groups states with the same observation

The signal distinguishes the two information sets

$$\{80\} \qquad \text{and} \qquad \{100, 130\}.$$

Hence

$$\sigma(Y) = \{\emptyset, \Omega, \{80\}, \{100, 130\}\}.$$

The call payoff takes values 0 and 30 inside $\{100, 130\}$, so it is **not** $\sigma(Y)$-measurable.

The digital payoff equals the signal:

$$\mathbf{1}_{\{S_T \geq 100\}} = Y,$$

so it is $\sigma(Y)$-measurable.

**Test**

The signal determines a quantity exactly if that quantity is constant on every information set.

# A conditional distribution updates the relevant probabilities

Return to the joint distribution

| | $Y = 0$ | $Y = 1$ |
|---|---|---|
| $X = 0$ | 0.30 | 0.10 |
| $X = 1$ | 0.20 | 0.40 |

Since $\mathbb{P}(Y = 1) = 0.50$,

$$\mathbb{P}(X = 1 \mid Y = 1) = \frac{\mathbb{P}(X = 1, Y = 1)}{\mathbb{P}(Y = 1)} = \frac{0.40}{0.50} = 0.80.$$

Observing $Y = 1$ replaces the marginal law of $X$ by the conditional law of $X$ given that information.

# Conditional expectation is the best prediction from current information

**Information interpretation**

$\mathbb{E}[X \mid \mathcal{G}]$ is the prediction of $X$ based only on the information in $\mathcal{G}$.

It must be $\mathcal{G}$-measurable because a prediction cannot use information that has not been observed.

Conditioning on a random variable is shorthand:

$$\mathbb{E}[X \mid Y] := \mathbb{E}[X \mid \sigma(Y)].$$

In a finite partition, the prediction is the average of $X$ inside the observed information set.

# Definition: conditional expectation matches averages on known events

## Conditional expectation

For $X \in L^1$ and $\mathcal{G} \subseteq \mathcal{F}$, $\mathbb{E}[X \mid \mathcal{G}]$ is the almost surely unique $\mathcal{G}$-measurable random variable $Z$ such that

$$\int_A Z \, d\mathbb{P} = \int_A X \, d\mathbb{P} \qquad \text{for every } A \in \mathcal{G}.$$

## Finite partition rule

On each information set $A$ with positive probability,

$$\mathbb{E}[X \mid \mathcal{G}](\omega) = \mathbb{E}[X \mid A], \qquad \omega \in A.$$

# Example A: compute a prediction from a partial signal

On $\Omega = \{1, 2, 3, 4\}$ with equal probabilities, let

$$X(1) = 0,\ X(2) = 4,\ X(3) = 8,\ X(4) = 12,$$

and let $Y = L$ on $\{1, 2\}$ and $Y = H$ on $\{3, 4\}$.

1. What partition and sigma-algebra does $Y$ reveal?
2. Which values can $\mathbb{E}[X \mid Y]$ take?
3. Average $X$ inside each information set.

Working together gives

$$\mathbb{E}[X \mid Y] = 2 \cdot \mathbf{1}_{\{Y=L\}} + 10 \cdot \mathbf{1}_{\{Y=H\}}.$$

Same information implies the same conditional prediction.

# Exercise B: value a payoff after a coarse signal

A fair die is rolled and

$$X = 12 \cdot \mathbf{1}_{\{\omega=6\}}.$$

You observe only whether

$$L = \{1, 2, 3, 4\} \qquad \text{or} \qquad H = \{5, 6\}$$

occurred.

1. Write $\sigma(L, H)$.
2. Compute $\mathbb{E}[X \mid L]$ and $\mathbb{E}[X \mid H]$.
3. Write the random variable $\mathbb{E}[X \mid \sigma(L, H)]$.
4. Check that its expectation equals $\mathbb{E}[X]$.

# Solution: average the payoff inside each information set

The revealed sigma-algebra is

$$\sigma(L, H) = \{\emptyset, L, H, \Omega\}.$$

On $L$, the payoff is always zero. On $H = \{5, 6\}$, its two values are 0 and 12. Therefore

$$\mathbb{E}[X \mid L] = 0, \qquad \mathbb{E}[X \mid H] = 6,$$

and

$$\mathbb{E}[X \mid \sigma(L, H)] = 6 \cdot \mathbf{1}_H.$$

Finally,

$$\mathbb{E}[\mathbb{E}[X \mid \sigma(L, H)]] = 6 \times \frac{2}{6} = 2 = \mathbb{E}[X].$$

# A joint density turns conditioning into a calculation

Suppose $(X, Y)$ has joint density $p_{X,Y}$. For $p_Y(y) > 0$,

$$p_Y(y) = \int_{\mathbb{R}} p_{X,Y}(x, y)\, dx,$$
$$p_{X|Y}(x \mid y) = \frac{p_{X,Y}(x, y)}{p_Y(y)},$$
$$g(y) = \int_{\mathbb{R}} f(x) p_{X|Y}(x \mid y)\, dx = \mathbb{E}[f(X) \mid Y = y].$$

Consequently,

$$\boxed{\mathbb{E}[f(X) \mid Y] = g(Y)}.$$

**Measurability check**

$g(Y)$ is a function of the observed signal $Y$, so it is $\sigma(Y)$-measurable.

# Worked example: conditioning with a joint density

Let

$$f_{X,Y}(x,y) = 2 \cdot \mathbf{1}_{\{0<x<y<1\}}.$$

For $0 < y < 1$, first compute the marginal density:

$$f_Y(y) = \int_0^y 2\,dx = 2y.$$

Then

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{1}{y}\mathbf{1}_{(0,y)}(x).$$

Therefore

$$\mathbb{E}[X \mid Y = y] = \int_0^y x\frac{1}{y}\,dx = \frac{y}{2}.$$

Thus

$$\mathbb{E}[X \mid Y] = \frac{Y}{2},$$

which is a function of $Y$.

# Student exercise: reverse the triangular density

Let

$$f_{X,Y}(x,y) = 2 \cdot \mathbf{1}_{\{0<y<x<1\}}.$$

For a fixed $0 < y < 1$:

1. Draw or describe the support of $(X, Y)$.
2. Compute $f_Y(y)$ by integrating out $x$.
3. Find $f_{X|Y}(x \mid y)$ and its support.
4. Compute $\mathbb{E}[X \mid Y = y]$.

**Hint**

For fixed $y$, the admissible values of $x$ run from $y$ to 1.

# Solution: conditioning on the reversed triangle

For fixed $0 < y < 1$,

$$p_Y(y) = \int_y^1 2\,dx = 2(1-y).$$

Hence

$$p_{X|Y}(x \mid y) = \frac{2}{2(1-y)}\mathbf{1}_{(y,1)}(x) = \frac{1}{1-y}\mathbf{1}_{(y,1)}(x).$$

Therefore

$$\mathbb{E}[X \mid Y = y] = \int_y^1 \frac{x}{1-y}\,dx = \frac{1+y}{2},$$

and, as a random variable,

$$\mathbb{E}[X \mid Y] = \frac{1+Y}{2}.$$

# Gaussian conditioning remains Gaussian

Suppose $(X, Y)$ is jointly Gaussian, with means $\mu_X, \mu_Y$, standard deviations $\sigma_X, \sigma_Y > 0$, and correlation $\rho$. Then

$$X \mid (Y = y) \sim \mathcal{N}\Big(\mu_X + \rho\frac{\sigma_X}{\sigma_Y}(y - \mu_Y),\ \sigma_X^2(1 - \rho^2)\Big).$$

Consequently,

$$\mathbb{E}[X \mid Y] = \mu_X + \rho\frac{\sigma_X}{\sigma_Y}(Y - \mu_Y), \qquad \mathrm{Var}(X \mid Y) = \sigma_X^2(1 - \rho^2).$$

**Why?**

With $a = \mathrm{Cov}(X, Y)/\mathrm{Var}(Y)$, the residual $R = X - \mu_X - a(Y - \mu_Y)$ is uncorrelated with $Y$. Joint Gaussianity makes $R$ independent of $Y$, which gives both formulas.

Larger $|\rho|$ means that observing $Y$ removes more uncertainty about $X$.

# Conditional expectation respects algebra and information

For integrable $X, Y$ and constants $a, b$:

**Linearity, order, and information**

$$\mathbb{E}[aX + bY \mid \mathcal{G}] = a\mathbb{E}[X \mid \mathcal{G}] + b\mathbb{E}[Y \mid \mathcal{G}],$$
$$X \leq Y \Longrightarrow \mathbb{E}[X \mid \mathcal{G}] \leq \mathbb{E}[Y \mid \mathcal{G}].$$

If $X$ is $\mathcal{G}$-measurable, then $\mathbb{E}[X \mid \mathcal{G}] = X$. If $X$ is independent of $\mathcal{G}$, then $\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X]$.

# Known factors come out and predictions satisfy the tower rule

Let $Z$ be $\mathcal{G}$-measurable with $ZX \in L^1$, and let $\mathcal{H} \subseteq \mathcal{G}$.

**Taking out what is known**

$$\mathbb{E}[ZX \mid \mathcal{G}] = Z\mathbb{E}[X \mid \mathcal{G}].$$

**Tower property**

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}] = \mathbb{E}[X \mid \mathcal{H}], \qquad \mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[X].$$

Re-predicting with less information is equivalent to predicting directly from that less detailed information.

Processes and martingales

Information evolving through time

Random walks and martingales

# A stochastic process is a collection of dated random variables

**Discrete-time stochastic process**

A process $(X_n)_{n\geq 0}$ is a sequence of random variables defined on the same probability space.

There are two complementary viewpoints:

- fix $n$: $X_n$ is one random variable with a distribution;
- fix $\omega$: $n \mapsto X_n(\omega)$ is one sample path.

The same tools now have a time index: distributions describe each date, joint laws describe dependence across dates, and information determines what can be predicted.

# Coin tosses show how information grows over time

Let $\xi_1, \xi_2$ be successive coin-toss outcomes and let $S_n$ count heads by time $n$.

| Time | Observed variables | Information available |
|---|---|---|
| 0 | none | no toss is known |
| 1 | $\xi_1$ | first toss is known |
| 2 | $\xi_1, \xi_2$ | full two-toss history is known |

The information sets become finer as observations arrive. A prediction at time 1 may use $\xi_1$, but not $\xi_2$.

# Filtrations record information and adapted processes respect it

## Filtration

A filtration $(\mathcal{F}_n)$ is an increasing family of sigma-algebras:

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots .$$

Here $\mathcal{F}_n$ is the information available up to time $n$.

## Adapted process

$(X_n)$ is adapted if $X_n$ is $\mathcal{F}_n$-measurable for every $n$.

Adapted means no look-ahead: today's value or decision can use the observed past and present, never the future.

# Student exercise: detect look-ahead

Two fair coin tosses are encoded by $\xi_1, \xi_2 \in \{0, 1\}$ and

$$\mathcal{F}_0 = \{\varnothing, \Omega\}, \qquad \mathcal{F}_1 = \sigma(\xi_1), \qquad \mathcal{F}_2 = \sigma(\xi_1, \xi_2).$$

Consider

$$X_0 = 0, \qquad X_1 = \xi_1 + \xi_2, \qquad X_2 = \xi_1 + \xi_2.$$

1. Is $(X_0, X_1, X_2)$ adapted to $(\mathcal{F}_n)$?
2. Replace $X_1$ by a natural adapted value while keeping $X_2$ unchanged.

## Solution: time 1 cannot use the second toss

$X_0$ is constant and hence $\mathcal{F}_0$-measurable. The terminal value $X_2 = \xi_1 + \xi_2$ is $\mathcal{F}_2$-measurable.

At time 1, however, $X_1$ depends on $\xi_2$, which is not yet observed. Thus $X_1$ is not $\mathcal{F}_1$-measurable and the process is not adapted.

The cumulative number of heads

$$\widetilde{X}_0 = 0, \qquad \widetilde{X}_1 = \xi_1, \qquad \widetilde{X}_2 = \xi_1 + \xi_2$$

is adapted.

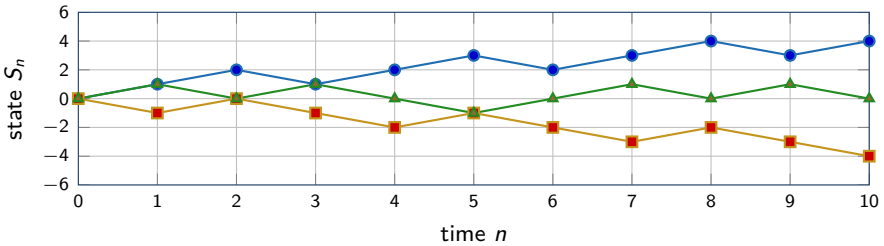
**Diagnostic**

At each date, check whether the value is determined by the information available at that date.

# A random walk evolves through many possible sample paths

Let $\varepsilon_k \in \{-1, +1\}$ and

$$S_n = S_0 + \sum_{k=1}^{n} \varepsilon_k.$$




One model generates many possible paths; only one is observed in a realized history.

# Independence makes random-walk moments tractable

Let the increments be i.i.d. with

$$\mathbb{E}[\varepsilon_k] = m, \qquad \mathrm{Var}(\varepsilon_k) = \sigma^2.$$

**Random-walk moments**

$$\mathbb{E}[S_n] = S_0 + nm, \qquad \mathrm{Var}(S_n) = n\sigma^2.$$

For a symmetric $\pm 1$ walk, $m = 0$ and $\sigma^2 = 1$. The mean stays at $S_0$ while uncertainty grows linearly with time.

# A martingale is a fair conditional prediction

**Information interpretation**

Given everything known today, the best prediction of tomorrow's value is today's value.

**Martingale**

An integrable adapted process $(M_n)$ is a martingale if

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n.$$

This is the mathematical form of a fair game: past wins and losses are known, but no predictable gain remains.

# Worked proof: the centered random walk is a martingale

Let

$$M_n = \sum_{k=1}^{n} \varepsilon_k,$$

where the increments are independent, centered, and $\mathcal{F}_n = \sigma(\varepsilon_1, \dots, \varepsilon_n)$.

Then $M_n$ is known at time $n$, while $\varepsilon_{n+1}$ is independent of $\mathcal{F}_n$:

$$\begin{aligned} \mathbb{E}[M_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}[M_n + \varepsilon_{n+1} \mid \mathcal{F}_n] \\ &= M_n + \mathbb{E}[\varepsilon_{n+1}] \\ &= M_n. \end{aligned}$$

This calculation uses measurability, independence, conditional expectation, and filtration in one place.

# Student exercise: find a second random-walk martingale

Let $S_n = \sum_{k=1}^{n} \varepsilon_k$, where the independent increments satisfy

$$\mathbb{P}(\varepsilon_k = 1) = \mathbb{P}(\varepsilon_k = -1) = \frac{1}{2}, \qquad \mathcal{F}_n = \sigma(\varepsilon_1, \ldots, \varepsilon_n).$$

Show that

$$M_n = S_n^2 - n$$

is a martingale.

**Hint**

Expand $S_{n+1}^2 = (S_n + \varepsilon_{n+1})^2$ and condition on $\mathcal{F}_n$.

# Solution: subtract the predictable variance growth

Since $S_n$ is $\mathcal{F}_n$-measurable and $\varepsilon_{n+1}$ is independent of $\mathcal{F}_n$,

$$\begin{aligned}\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}[(S_n + \varepsilon_{n+1})^2 - (n+1) \mid \mathcal{F}_n] \\ &= S_n^2 + 2S_n\mathbb{E}[\varepsilon_{n+1}] + \mathbb{E}[\varepsilon_{n+1}^2] - (n+1) \\ &= S_n^2 - n = M_n.\end{aligned}$$

**Interpretation**

$S_n^2$ grows predictably by one unit per step; subtracting $n$ removes that drift.

# Martingales connect fair games to financial pricing

- In a fair casino game, the conditional expected future wealth equals current wealth.
- Under a risk-neutral measure $\mathbb{Q}$, discounted traded asset prices are martingales.
- Therefore a replicable payoff satisfies

$$\frac{V_t}{B_t} = \mathbb{E}^{\mathbb{Q}}\left[\frac{X}{B_T}\,\middle|\,\mathcal{F}_t\right].$$

Replication supplies the economics; the martingale property supplies the dynamic conditional-expectation formula.

Short differential-calculus review

Derivatives needed later in the course

# The derivative is defined by a limiting difference quotient

**Derivative of $g : \mathbb{R} \to \mathbb{R}$**

If the limit exists, the derivative of $g$ at $x$ is

$$g'(x) = \lim_{h \to 0} \frac{g(x+h) - g(x)}{h}.$$

It measures the first-order change:

$$g(x+h) = g(x) + g'(x)h + o(h).$$

For example, applying the definition to $g(x) = x^2$ gives $g'(x) = 2x$.

# Partial derivatives vary one coordinate at a time

Let $f : \mathbb{R}^d \to \mathbb{R}$ and let $e_i$ be the $i$th coordinate vector.

## Partial derivative and gradient

$$\partial_i f(x) = \lim_{h \to 0} \frac{f(x + h e_i) - f(x)}{h}, \qquad \nabla f(x) = \big(\partial_1 f(x), \ldots, \partial_d f(x)\big).$$

If the second partial derivatives exist, the Hessian is

## Hessian

$$\nabla^2 f(x) = \left( \frac{\partial^2 f}{\partial x_i \partial x_j}(x) \right)_{1 \le i,j \le d}.$$

The gradient gives first-order sensitivities; the Hessian records curvature and interactions.

# Worked example: compute the gradient and Hessian

For

$$f(x, y) = x^2 y + e^y,$$

the first partial derivatives are

$$f_x = 2xy, \qquad f_y = x^2 + e^y, \qquad \nabla f(x, y) = (2xy, x^2 + e^y).$$

Differentiating once more gives

$$f_{xx} = 2y, \qquad f_{xy} = f_{yx} = 2x, \qquad f_{yy} = e^y,$$

and therefore

$$\nabla^2 f(x, y) = \begin{pmatrix} 2y & 2x \\ 2x & e^y \end{pmatrix}.$$

# The chain rule connects sensitivities to value changes

If $x(t) \in \mathbb{R}^d$ is differentiable and $f$ is differentiable, then

**Multivariable chain rule**

$$\frac{d}{dt} f(x(t)) = \nabla f(x(t)) \cdot x'(t).$$

For an option value $V = V(t, S)$ and a differentiable price path $S = S(t)$,

$$\frac{d}{dt} V(t, S(t)) = V_t(t, S(t)) + V_S(t, S(t)) S'(t).$$

Here $V_S$ measures sensitivity to the underlying price, $V_t$ measures sensitivity to calendar time, and $V_{SS}$ measures curvature in the underlying price.